



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems

Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech CSBS

Course Code & Title : CS3391 & Object Oriented Programming

Name of the Faculty member(s): Dr.S.Erana Veerappa Dinesh

Innovative Practice Description

- **Unit/Topic:** Unit II / Overloading Methods

- **Course Outcome:**CO2

- **Topic Learning Outcome:**TLO6

- **Activity Chosen:** Virtual Lab

- **Justification:**

The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is a non screen simulator that learners use to test ideas and observe results. Since the method overloading is mandatory for the java programming language, hands on demonstration is given to the students through Virtual lab by using the following link

<https://java-iitd.vlabs.ac.in/>

- **Time Allotted for the Activity :** 20 Minutes

- **Details of the Implementation:**

The course instructor explained the concept of method creation and overloading the methods in java. Figure 1 and 2 shows the Procedure for implementing the method overloading in java. He opened the Virtual Lab using a fore mentioned link and simulates the method overloading in java to the students. At last 10 minutes, the course instructor chosen few students and instructed them to execute and explain the examples in virtual lab by the students shown in the fig 3.

- **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PSO1
CO1	2	2	2	1

(1– Low

2–Moderate

3 – High)

- PO/ PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO01
	2	2	1	2
Justification for correlation	Students could be able to apply basic syntax and principles of Java methods	Students could be able to have the hands-on with implementing method overloading on Virtual lab	Students could be able to recollect the concept of overloading methods in java	Students could be able to knowledge about the method overloading to their peer group of people.

- Images/Screenshot of the practice:

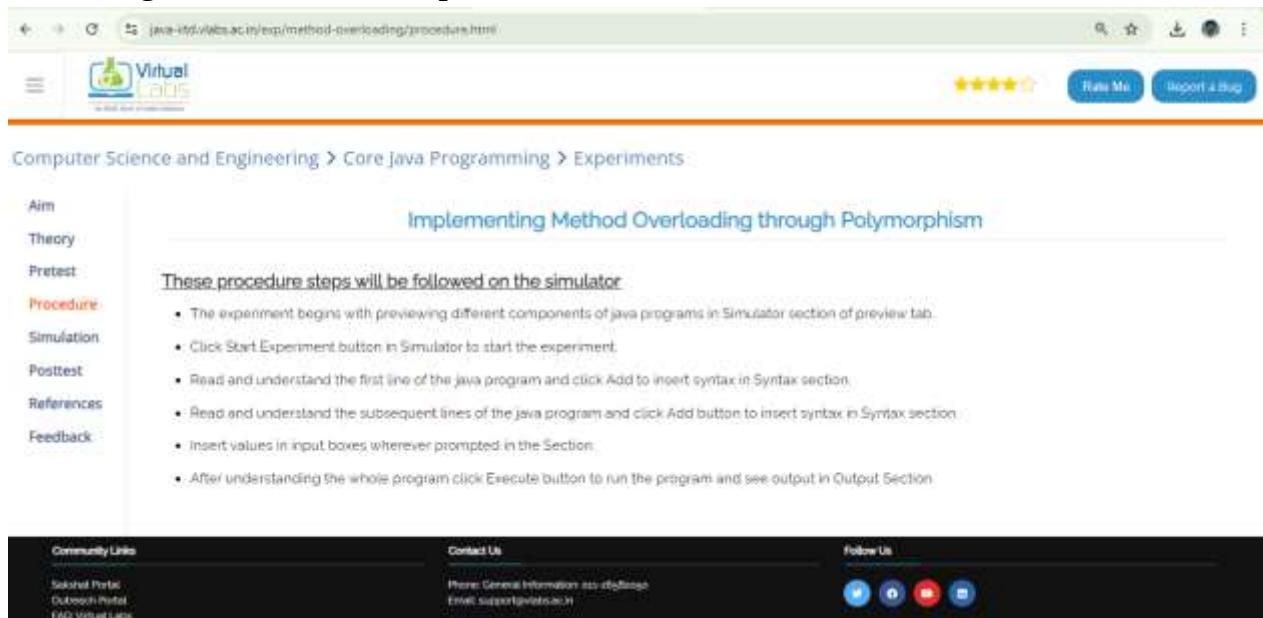


Fig1: Procedure for implementing method overloading in java

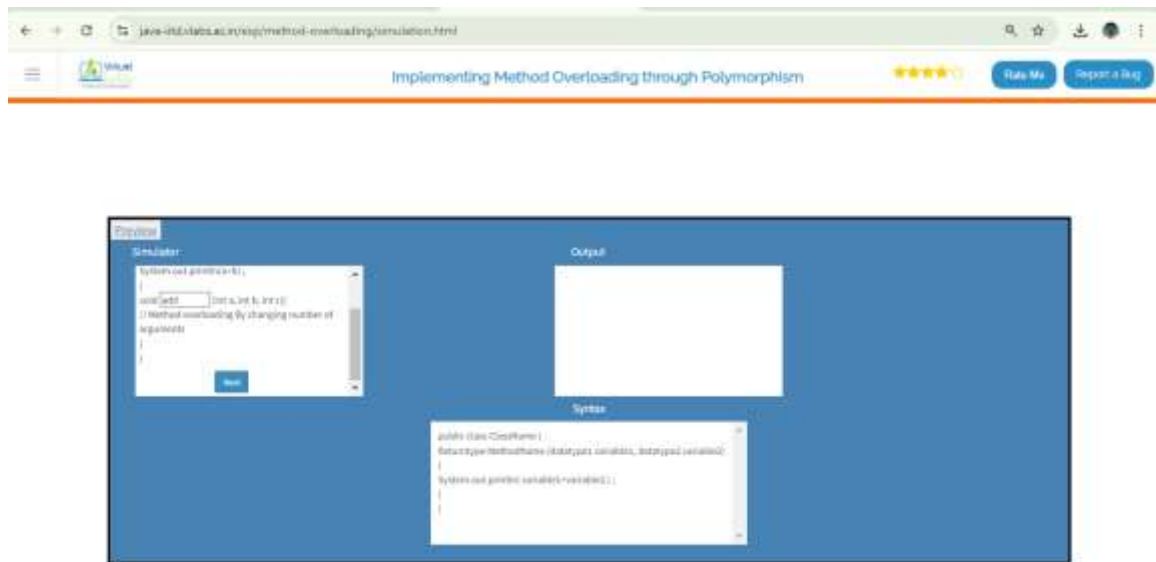


Fig:2 Simulation on method overloading

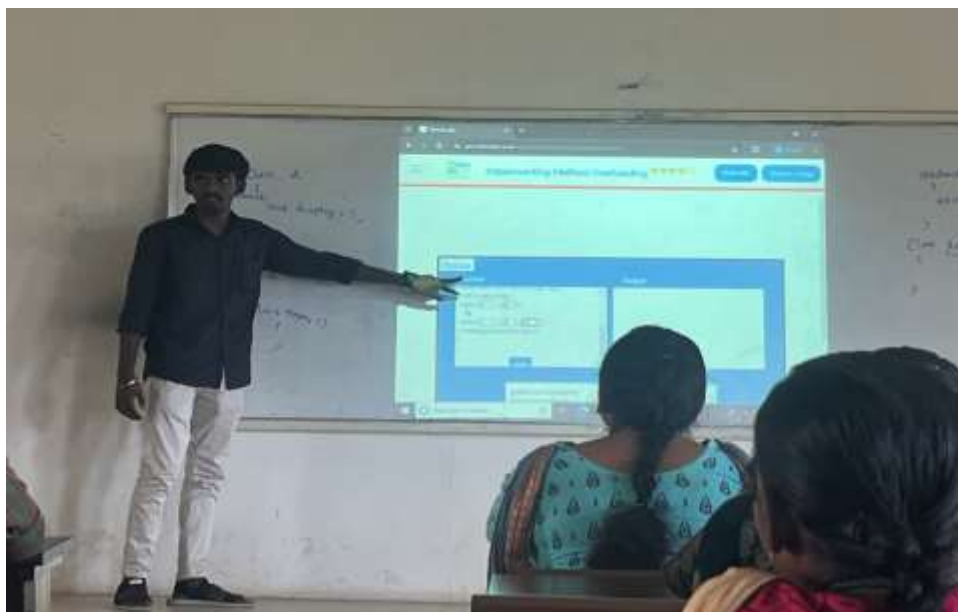


Fig:3 Demonstration by the student Santhosh

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students easily understand the creation of classes, objects and its corresponding code.
 - Students said that the pretest and posttest is very useful to recollect the concepts.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students had understood the concepts of the classes and objects in java.
 - Most of the students were actively participated and enjoyed once they answered correctly in pretest.

❖ ***Challenges faced in implementation:***

- Some students only involved in the demonstration activity.
- The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VLAB_AL3452_OS_MGM_23_24_EVEN.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VLab_GE3151_PSPP_KU_23_24.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2022-2023/SEV_GE3151_VLab.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/VL_GE3151_PSPP_BY_23_24.pdf